STAT340 Lecture: Resampling and the Bootstrap

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Introduction

In previous lectures throughout the semester, we have looked at different ways of quantifying uncertainty in our data and our resulting estimates. Among the most fundamental tools in statistics for quantifying uncertainty is the bootstrap.

Ultimately, the bootstrap amounts to resampling our data as though it were the population itself.

Rather surprisingly, this can actually help us estimate certain quantities related to variances (i.e., uncertainty).

Learning objectives

After this lecture, you will be able to

- Explain the bootstrap and its applicability.
- Implement and apply the bootstrap to estimate variance in simple models.

Estimating Variance

In a wide range of applications, we need to estimate variance of a population from which we sample data.

Example: confidence intervals

Given data $X_1, X_2, \ldots, X_n$ drawn from some distribution, we are often interested in constructing a confidence interval for some parameter θ. For example, θ might be the mean of our distribution. If an estimator $\hat{\theta}$ is approximately normal about θ, we could construct an approximate CI for θ if only we knew the standard deviation of $\hat{\theta}$,

$$\sigma_{\hat{\theta}} = \sqrt{\mathrm{Var}\,\hat{\theta}}.$$

Example: errors in linear regression

Recall that in linear regression we observe (X_i, Y_i) pairs where

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, 2, \ldots, n$$

where $\epsilon_1, \epsilon_2, \ldots, \epsilon_n$ are drawn iid according to a normal with mean zero and unknown variance $\sigma^2 > 0$.

Often we were interested in the sampling distribution of the estimated coefficients $\hat{\beta}_0, \hat{\beta}_1$

Often, we need to know σ^2 or at least have a good estimate of it. How might we estimate σ^2?

Tricky variance computations

In situations like the above, we had central limit theorems or similar results that allowed us to estimate the variance of the quantities we cared about. Unfortunately, this isn't always the case. Here's a simple example, adapted from ISLR (beginning of Section 5.2):

Example: allocating investments

Suppose that we have two different stocks we can invest in, which yield returns X and Y. So, we invest a single dollar (just to keep things simple!), and we need to split between these two different stocks. Let's say we put α dollars into the first stock (with return X) and $1 - \alpha$ (i.e., the rest) into the second stock (with return Y).

Then we get back an amount $\alpha X + (1 - \alpha)Y$.

Now, this amount of money that we get back on our investment is random, because it is based on the (random) stock yields X and Y. An obvious thing to do is to try and maximize this, but this can result in us making risky bets.

Minimizing variance

Most investment strategies instead aim to minimize the "risk" (In finance, "risk" means "volitility" i.e. variance):

$$\begin{aligned}\operatorname{Var}(\alpha X + (1-\alpha)Y) &= \alpha^2 \operatorname{Var} X + (1-\alpha)^2 \operatorname{Var} Y + 2\alpha(1-\alpha)\operatorname{Cov}(X, Y) \\ &= \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}\end{aligned}$$

Where

$$\begin{aligned}\sigma_X^2 &= \operatorname{Var} X \\ \sigma_Y^2 &= \operatorname{Var} Y \\ \sigma_{XY} &= \operatorname{Cov}(X, Y).\end{aligned}$$

One can prove that $\operatorname{Var}(\alpha X + (1-\alpha)Y)$ is minimized by taking

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Unfortunately, we don't know the variance and covariance terms (i.e., σ_X^2, σ_Y^2, σ_{XY})...

Estimating the parameters

But suppose we have observations (X_i, Y_i), $i = 1, 2, \ldots, n$ of, say, the performance of these two stocks on prior days. We could use this data to estimate σ_X^2, σ_Y^2 and σ_{XY}.

If we denote the estimates $\hat{\sigma}_X^2$, $\hat{\sigma}_Y^2$ and $\hat{\sigma}_{XY}$, then we could plug these into the above equation and obtain an estimator for α,

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

Quantifying uncertainty

What if we want to quantify our uncertainty about α?

After all, $\hat{\alpha}$ is a function of the (random) data, so it is itself a random variable, and it is reasonable to ask about its variance.

But $\hat{\alpha}$ depends on the data $\{(X_i, Y_i) : i = 1, 2, \ldots, n\}$ in a fairly complicated way, so it's not obvious what $\operatorname{Var} \hat{\alpha}$ should actually be.

Refresher: simulation-based methods

In our discussion of estimation and confidence intervals earlier in the semester, we obtained simulation-based confidence intervals by

1. Assuming a model for our data (e.g., assuming the data came from a Poisson distribution)
2. Estimating the parameter(s) of that model from the data (e.g., estimating λ under the Poisson)
3. Generating new random samples from the model with the estimated parameter(s) (e.g., drawing from $\text{Pois}(\hat{\lambda})$)
4. Using these "fake" data samples to approximate the sampling distribution of our statistic of interest (e.g., $\hat{\lambda}(X_1, X_2, \ldots, X_n)$).

If we had a model

If we had a model for our (X_i, Y_i) pairs, we could use the observed pairs to estimate the parameter(s) of that model and generate new samples $(X_i', Y_i'), i = 1, 2, \ldots, n$ and compute

$$\hat{\alpha}' = \hat{\alpha}\left((X_1', Y_1'), (X_2', Y_2'), \ldots, (X_n', Y_n')\right).$$

Doing this many times, we would get multiple random variables that approximate the distribution of $\hat{\alpha}$ (i.e., the statistic computed on our original data).

What if we don't have a model?

Simulation-based approaches like the one discussed above work because we have made a model assumption about our data. If we assumed our data came from a Poisson distribution, then we could just estimate the Poisson parameters and generate new samples.

But often we don't want to make such assumptions about our data. Because, for example

- Our parameter(s) may be expensive to estimate
- The distribution may be expensive to draw from
- We don't want to make model assumptions in the first place!

This last concern gives rise to what we call *non-parametric statistics*. That is, doing statistics while avoiding assumptions of the form "We assume that the data are generated according to a normal (or Poisson or Binomial or. . .)".

Introducing the Bootstrap

What we want to do is to draw new samples from the same population (i.e., distribution) as our data came from. In the Monte Carlo simulation-based approach, we estimate the parameter(s) to get an approximation to that true distribution, and we use this model to simulate data.

The **bootstrap** takes a different approach. The data, $X_1, X_2, \ldots, X_n$ is a sample from the *actual* population that we care about (i.e., not an approximation!). The bootstrap says, "let's just sample from $X_1, X_2, \ldots, X_n$."

Resampling with Replacement

Said another way, in the bootstrap, we sample *with replacement* from the observed data

$$X_1, X_2, \ldots, X_n,$$

obtaining the sample

$$X_1^*, X_2^*, \ldots, X_n^*.$$

The X_i^* notation is convention in statistics– the asterisk $(*)$ denotes that the variable is resampled from the original data.

Important note: X_i^* does *not* necessarily equal X_i. It is just the i-th resampled data point. X_i^* is equal to *some* X_j, $j = 1, 2, \ldots, n$, with each of the n different data points being equally likely. Each X_i^* is a sample with replacement from the data $X_1, X_2, \ldots, X_n$.

A bootstrapped estimator

Now, having resampled our data, we can compute our estimator on the bootstrap sample $X_1^*, X_2^*, \ldots, X_n^*$, say,

$$\hat{\theta}\left(X_1^*, X_2^*, \ldots, X_n^*\right).$$

Repeating this many times (say, B times), we can use these resampled replicates of our original estimator, say

$$\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_B.$$

The intuition is that these B random variables, each based on a sample with replacement from the original data, are a good approximation to the true distribution of our estimate $\hat{\theta}(X_1, X_2, \ldots, X_n)$, and we can use them to do things like estimating the variance.

And it works?

Yes, it works (for certain problems, anyway).

The intuition is something like this: when n is suitably large, the "point cloud" formed by $X_1, X_2, \ldots, X_n$ looks a lot like the true population distribution.

As a result, resampling from the observed sample $X_1, X_2, \ldots, X_n$ and resampling from the true population distribution are not actually so different!

Example: estimating the Poisson rate parameter

Let's suppose that $X_1, X_2, \ldots, X_n$ are drawn iid from a Poisson distribution with rate parameter $\lambda = 5$.

Our goal is to construct a confidence interval for the rate parameter λ.

While other methods are better here, it will serve as an example of how bootstrap estimation is done.

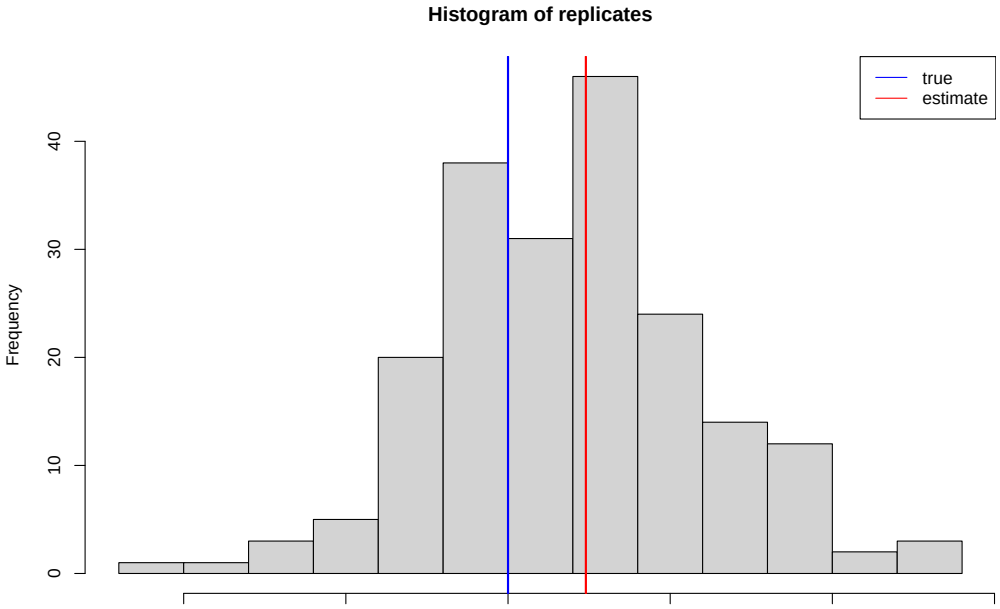
```
n = 25
lambdatrue = 5
data = rpois(n=n, lambda=5); # Generate a sample of 25 iid RVs
lambdahat = mean(data); # Estimate lambda.
```

Now, let's do the bootstrap. We'll repeatedly (B times) resample n=25 observations from the data. On each resample, compute lambdahat (i.e., take the mean).

```
B = 200; # Number of bootstrap replicates.
replicates = rep(NA,B); # We'll store estimates here
for( i in 1:B ) {
  # Sample WITH REPLACEMENT from the data sample itself.
  resample = sample(data, n, replace=TRUE)
  # Compute our statistic on the resample data.
  # This is a *bootstrap replicate* of our statistic.
  replicates[i] = mean(resample)
}
```

Now, `replicates` is a vector of (approximate) replicates of our estimate for λ.

Bootstrap distribution



Constructing the Interval

While there are better methods, we'll look at a simple way to produce a confidence interval for illustration purposes.

Use those bootstrap replicates to estimate the standard deviation of $\hat{\lambda} = \frac{1}{n}\sum_i X_i$, and then use that to get a 95% CI for λ.

```
sd_lambda = sd( replicates )
CI = c( lambdahat-1.96*sd_lambda, lambdahat+1.96*sd_lambda)
# And check if our CI contains lambda = 5.
(CI[1] < lambdatrue) & (lambdatrue < CI[2])
```

```
## [1] TRUE
```

Well, about 95% of the time, we should catch $\lambda = 5$.

Verifying Bootstrap accuracy (coverage)

Let's run that same experiment a bunch of times, just to verify. That is, we are going to:

1. Generate data $X_1, X_2, \ldots, X_n$ iid from Poisson with $\lambda = 5$.
2. Compute $\hat{\lambda} = \bar{X}$
3. Repeatedly resample from $X_1, X_2, \ldots, X_n$ *with replacement*, compute the mean of each resample (i.e., compute our estimator on each resample)
4. Compute the standard deviation of these resampled copies of $\hat{\lambda}$.
5. Use that SD to compute a 95% CI, under the assumption that $\hat{\lambda}$ is approximately normal about its expectation λ.

```r
run_pois_bootstrap_expt = function( lambdatrue, n, B ) {
  # lambdatrue is the true value of lambda to use in generating our data.
  # n is the sample size, B is the number of bootstrap replicates to use.
  data = rpois( n=n, lambda=lambdatrue ) #generate data
  lambdahat = mean( data ); # Our point estimate for lambda.
  replicates = rep(NA,B)
  # Each replicate draws n data points, with replacement, from data
  # and computes its mean (i.e., estimates lambda from the resampled data)
  for( i in 1:B ) {
    # resample n elements of the data, with replacement.
    resampled_data = sample( data, n, replace=TRUE )
    replicates[i] = mean( resampled_data )
  }
  # Now use those replicates to estimate SD of and construct a 95% CI.
  sd_boot = sd( replicates ); # estimate of the std dev of alphahat
  CI = c(lambdahat-1.96*sd_boot, lambdahat+1.96*sd_boot)
  # Finally, check if this CI caught lambda successfully.
  # Return TRUE/FALSE accordingly.
  return( (CI[1] < lambdatrue) & (lambdatrue < CI[2]) )
}
```

Run the experiment

Now, let's repeat this experiment a few hundred times, and see how often our CI contains the true λ.

```
set.seed(1)
N_expt = 500; # number of CIs to construct.
successes = rep(NA, N_expt); # Keep track of whether or not each CI caught alpha.
for( i in 1:N_expt ) {
  # Repeat the experiment N_expt times.
  # lambdatrue=5, B=200 bootstrap replicates, n=25 observations in the data sampl
  successes[i] = run_pois_bootstrap_expt( 5, 200, 25 )
}
mean(successes)
```

```
## [1] 0.946
```

Recap: general recipe for the bootstrap

We have data $X_1, X_2, \ldots, X_n$ drawn iid from some unknown distribution. We wish to estimate a parameter θ using estimator $\hat{\theta} = \hat{\theta}(X_1, X_2, \ldots, X_n)$.

1. Repeatedly (B times) sample n observations *with replacement* from $X_1, X_2, \ldots, X_n$, to obtain samples $X^*_{b,1}, X^*_{b,2}, \ldots, X^*_{b,n}$, for $b = 1, 2, \ldots, B$.
2. For each of these B resamples, compute the estimator on that sample, to obtain

$$\hat{\theta}_b = \hat{\theta}(X^*_{b,1}, X^*_{b,2}, \ldots, X^*_{b,n}) \quad \text{for } b = 1, 2, \ldots, B.$$

3. Compute the standard deviation of our bootstrap replicates,

$$\mathrm{SE}_{\mathrm{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\theta}_b - \frac{1}{B} \sum_{r=1}^{B} \hat{\theta}_r \right)^2}.$$

4. Construct an (approximate) confidence interval, under the assumption that $\hat{\theta}$ is normally distributed about its mean θ.

$$(\hat{\theta} - 1.96\,\mathrm{SE}_{\mathrm{boot}}, \hat{\theta} + 1.96\,\mathrm{SE}_{\mathrm{boot}})$$

Bootstrapping for more complicated statistics

Let's return to our example of financial returns (X_i, Y_i)

The bootstrap says that we should sample *with replacement* from the observed data

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n),$$

obtaining the sample

$$(X_1^*, Y_1^*), (X_2^*, Y_2^*), \ldots, (X_n^*, Y_n^*).$$

Again, the X_i^* notation is the conventional way to show that the variable is resampled from the original data.

Having resampled from our data in this way, we could then compute

$$\alpha^* = \hat{\alpha}\left((X_1^*, Y_1^*), (X_2^*, Y_2^*), \ldots, (X_n^*, Y_n^*)\right).$$

Repeating this, say, B times, we would obtain *bootstrap replicates*

$$\alpha_1^*, \alpha_2^*, \ldots, \alpha_B^*,$$

from which we can estimate the variance of $\hat{\alpha}$ as

$$\hat{\sigma}_\alpha^2 = \frac{1}{B-1} \sum_{r=1}^{B} \left(\alpha_r^* - \frac{1}{B} \sum_{b=1}^{B} \alpha_b^* \right)^2.$$

Notice that this *looks* like a variance, except that we are using $\hat{\alpha}$, the statistic computed on the original data sample, as our estimate of the mean.

Demo: applying the bootstrap to financial data

The following chunk contains code for generating (X_i, Y_i) pairs like those discussed in our financial example above.

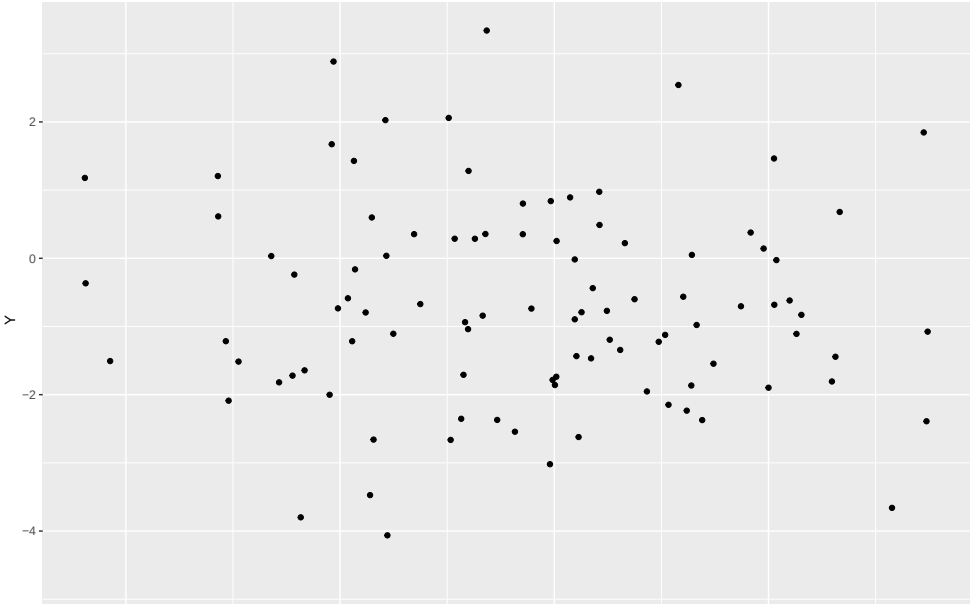
```
require(MASS)

generate_pairs = function( n ) {
  # Generate n pairs of financial returns.
  muX = 2; muY = -1
  CovMx = matrix( c(1,-.25,-.25,2), nrow = 2)
  data = mvrnorm(n=100, mu=c(muX,muY), Sigma=CovMx)
  return( data.frame( 'X'=data[,1], 'Y'=data[,2]) )
}
```

```r
fin_pairs = generate_pairs( 100 ); # Generate 100 (X,Y) pairs.
head(fin_pairs)
```

```
##             X          Y
## 1  2.25895101 -1.1941734
## 2  2.12729092 -0.7908007
## 3  3.03691780 -0.0264181
## 4  1.51629647 -2.6633349
## 5 -0.07325971 -1.5070794
## 6  2.53320235 -2.1479554
```

A scatterplot



Compute the estimate

Now, let's compute $\hat{\alpha}$ on this data. Remember, $\hat{\alpha}$ is just a function of the (estimated) variances of X and Y, along with their covariance:

$$\hat{\alpha} = \frac{\hat{\sigma}_Y^2 - \hat{\sigma}_{XY}}{\hat{\sigma}_X^2 + \hat{\sigma}_Y^2 - 2\hat{\sigma}_{XY}}.$$

So, let's just compute those three quantities and plug them in. The `cov` function gets us the whole sample covariance matrix of our data:

```
Sigmahat = cov( fin_pairs ); #Sigma is the common symbol for a covariance matrix.
Sigmahat
```

```
##             X          Y
## X  0.8304713 -0.1144192
## Y -0.1144192  2.3422483
```

Now we can just pluck our three estimates out of there

```
sigma2hatXX = Sigmahat[1,1]
sigma2hatYY = Sigmahat[2,2]
sigmahatXY = Sigmahat[1,2]
```

and we can plug these into our formula for $\hat{\alpha}$ above.

```
alphahat = (sigma2hatYY - sigmahatXY)/(sigma2hatXX + sigma2hatYY -2*sigmahatXY)
alphahat
```

```
## [1] 0.7222183
```

In truth, the covariances that generated our data are:

$$\sigma_X^2 = 1, \quad \sigma_Y^2 = 2, \quad \sigma_{XY} = -0.25,$$

so the true optimal choice of α is

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}} = \frac{2 - (-0.25)}{1 + 2 + 2 * 0.25} = \frac{2.25}{3.5} \approx 0.64$$

Let's store the true value of alpha while we're thinking of it.

```
sigma2XX = 1
sigma2YY = 2
sigmaXY = -0.25
alpha_true =(sigma2YY - sigmaXY)/(sigma2XX + sigma2YY -2*sigmaXY)
alpha_true
```

```
## [1] 0.6428571
```

Bootstrapping

Now, again, we're going to resample *with replacement* from our data, and compute our statistic $\hat{\alpha}$ on each resample. The hope is that these resampled versions of the statistic will resemble the distribution of the statistic evaluated on the original data.

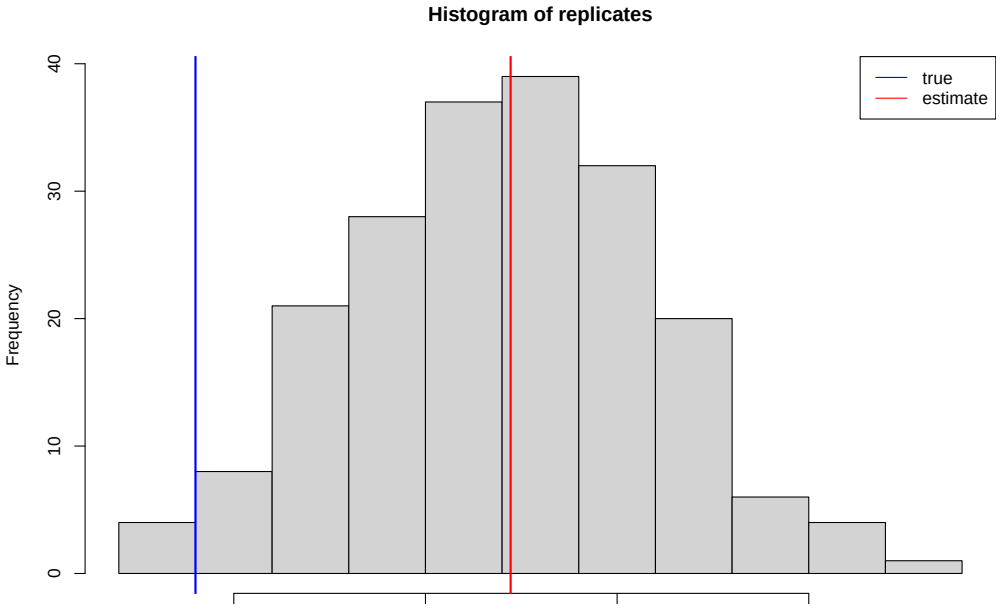
It will be convenient to just have a function to compute $\hat{\alpha}$ from a given data set.

```
compute_alphahat = function( data ) {
  # We're assuming that data is a data frame with two columns.
  Sigmahat = cov( data )
  # Extract the variance and covariance estimates from the sample covariance
  sigma2hatXX = Sigmahat[1,1]
  sigma2hatYY = Sigmahat[2,2]
  sigmahatXY = Sigmahat[1,2]
  # plug these into the definition of alpha.
  alphahat = (sigma2hatYY - sigmahatXY)/(sigma2hatXX + sigma2hatYY -2*sigmahatXY)
  return(alphahat)
}
alphahat = compute_alphahat( fin_pairs )
```

Okay, we're ready to go. Let's resample the data $B = 200$ times, evaluating $\hat{\alpha}$ on each resample. Then, we'll use those resampled values to estimate the variance.

```r
B = 200
replicates = rep(NA,B)
n = nrow( fin_pairs ); # number of observations in our data set.
for( i in 1:B ) {
  # To resample the data, we will sample indices, and then grab those rows.
  resample_indices = sample( 1:n, n, replace=TRUE )
  resampled_data = fin_pairs[resample_indices,]
  replicates[i] = compute_alphahat( resampled_data )
}
```

Visualizing the alpha hats



Estimating Variance

```r
# Estimate the variance of alphahat from our bootstrap replicates.
sd_alphahat = sd( replicates ); # estimate of the std dev of alphahat
CI = c(alphahat-1.96*sd_alphahat, alphahat+1.96*sd_alphahat)
CI
```

```
## [1] 0.6468077 0.7976289
```

. . . and we can verify that the CI contains the true value of α (about 95% of the time you run this code, anyway):

```r
(CI[1] < alpha_true) & (alpha_true < CI[2])
```

```
## [1] FALSE
```

Verify accuracy of the method

Just to verify, let's rerun this whole machinery a few times, and check that our bootstrap-based CI catches α about 95% of the time.

Importantly, that "95% of the time" means "if we generate new data, then 95% of the time, our bootstrap-based CI will catch the true value of α."

To do this experiment, we want a function to run the above procedure once for us on a given data frame and a given number of bootstrap replicates.

Running experiment 200 times

```r
set.seed(1)
N_expt = 200; # number of CIs to construct.
successes = rep(NA,N_expt); # Keep track of whether or not each CI caught alpha.
for( i in 1:N_expt ) {
  successes[i] = run_bootstrap_expt( 200 ); # Using B=200 replicates.
}
mean(successes)
```

```
## [1] 0.94
```

Limitations and extensions of the bootstrap

Here's a simple example of where the bootstrap can fail.

Example: estimating the distribution of the maximum

Suppose that we see data $X_1, X_2, \ldots, X_n$ drawn iid from some distribution, and we are interested in the distribution of

$$M = \max\{X_1, X_2, \ldots, X_n\}.$$

In particular, suppose we want to estimate $\mathbb{E}M$.

The bootstrap would suggest that we sample $X_1^*, X_2,^*, \ldots, X_n^*$ with replacement from $X_1, X_2, \ldots, X_n$ and compute

$$M^* = \max\{X_1^*, X_2^*, \ldots, X_n^*\}.$$

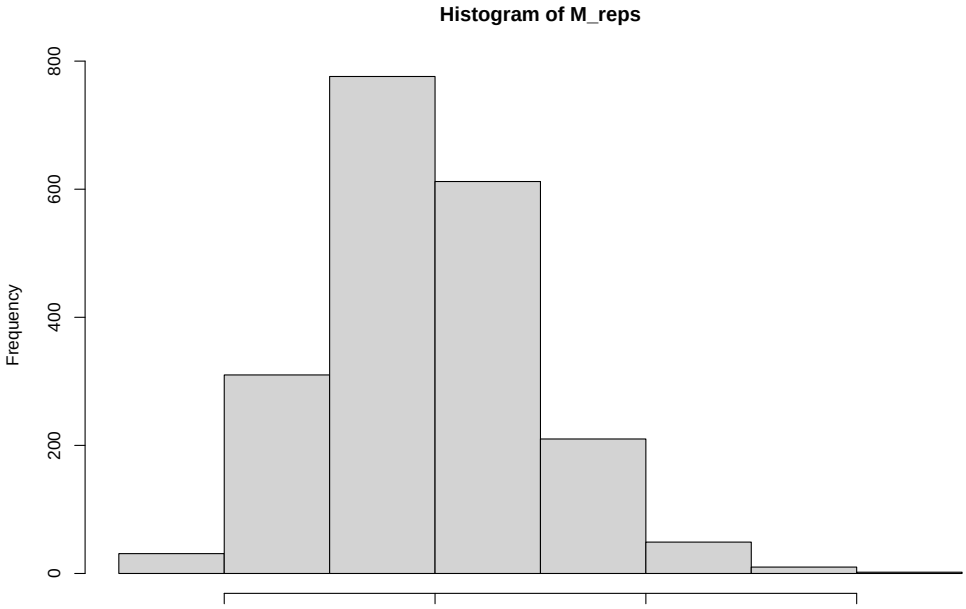
Repeating this B times, we get replicates $M_1^*, M_2^*, \ldots, M_B^*$, and we can compute their standard deviation to estimate the standard deviation of our variable of interest M.

Now let's implement that and see how well it works.

We'll draw our data from a standard normal. First, let's generate a bunch of copies of M, so we can see its distribution (and its variance, in particular).

```r
n_samp = 25; # Number of data samples to take
n_rep = 2000; # Number of replicates of M to create.
M_reps = rep( NA, n_rep )
for( i in 1:n_rep ) {
  data = rnorm(n=n_samp)
  M_reps[i] = max(data)
}
```

Histogram of M_reps



True value of $E(M)$

The true value of $\mathbb{E}M$ is actually very hard to compute, but we can use Monte Carlo estimation to approximate it:

```
ExpecM = mean(M_reps)
ExpecM
```

```
## [1] 1.962873
```

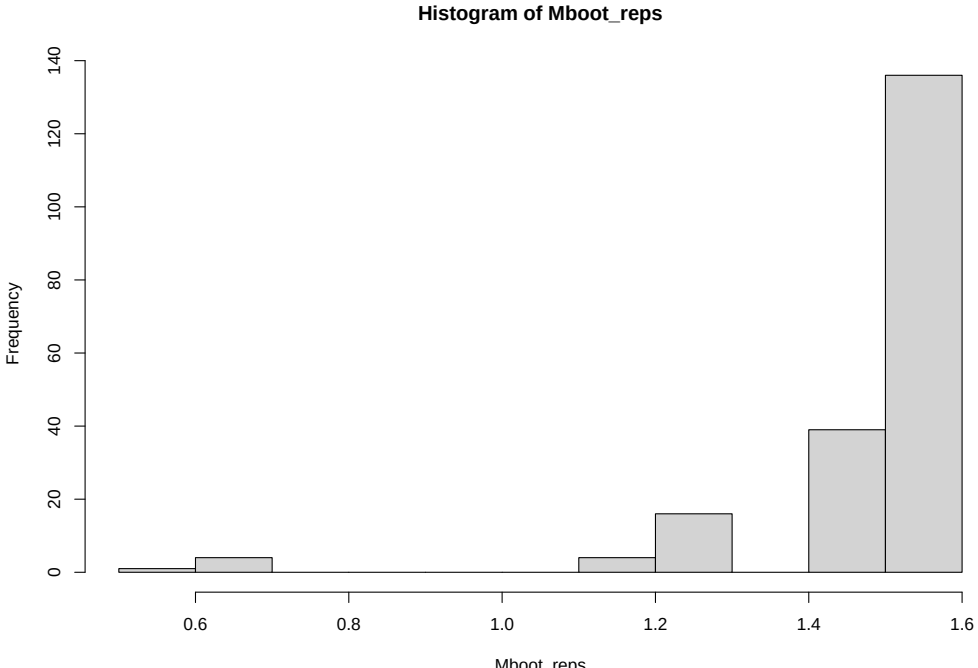
In particular, the (sample) standard deviation is

```
sd(M_reps)
```

```
## [1] 0.5015513
```

Now, let's generate a new sample and run the bootstrap on it.

```
data = rnorm(n=n_samp)
B = 200
Mboot_reps = rep( NA, B )
for( b in 1:B ){
  resamp = sample( data, n_samp, replace=TRUE)
  Mboot_reps[b] = max(resamp)
}
```


Histogram of Mboot_reps
Frequency
0
20
40
60
80
100
120
140
0.6
0.8
1.0
1.2
1.4
1.6
Mboot_reps

Estimating Variance

Let's just check the variance? Maybe it's close to the observed $\sigma_M \approx 0.483$?

```
sd(Mboot_reps)
```

```
## [1] 0.1527378
```

Compare with

```
sd(M_reps)
```

```
## [1] 0.5015513
```

And if we try to construct a confidence interval, we get a CI

```
M = max(data)
seMboot = sd(Mboot_reps)
CI = c( M-1.96*seMboot, M+1.96*seMboot )
CI
```

```
## [1] 1.204802 1.803535
```

Checking accuracy

Okay, but we really have to run the experiment multiple times, so let's package it up in a function. . .

```
# A function to repeat the same experiment easily without having to repeat oursel
run_max_boot_expt = function( n, B ) {
  data = rnorm(n)
  M = max(data)
  Mboot_reps = rep(NA,B)
  for(b in 1:B) {
    resamp = sample( data, n, replace=TRUE)
    Mboot_reps[b] = max(resamp)
  }
  seMboot = sd(Mboot_reps)
  CI = c( M-1.96*seMboot, M+1.96*seMboot )
  return( CI[1] < ExpecM & ExpecM < CI[2] )
}
```

. . . and, just like we've done above, we'll generate data samples a bunch of times and see how often our bootstrap CI catches the (estimated!) true value of $\mathbb{E}M$.

```
set.seed(1)
Nexpt = 500
successes = rep(NA,Nexpt)
for(i in 1:Nexpt) {
  successes[i] = run_max_boot_expt( n_samp, 200 )
}
mean(successes)
```

```
## [1] 0.724
```

Hmm. . . Not ideal.

The problem is that the bootstrap tends to vastly underestimate the variance of "challenging" functions like the maximum.

Generally speaking, the bootstrap only works well when the statistic we are trying to work with is approximately normal about its expectation. The maximum of a collection of variables does obey a central limit theorem, but its limiting distribution is *not* a normal!

When is Bootstrapping useful

Bootstrapping can be useful when

- trying to calculate confidence intervals for quantities whose standard deviations are not known or difficult to calculate (no analytical or closed-form solution)
- in estimating the accuracy of a predictive model
- If we do not wish to make any model assumptions about a population

Review

In these notes we covered

- A general method for bootstrap estimation
- Bootstrap estimation for univariate data
- Bootstrap estimation for bivariate data
- Limitations of bootstrapping